

ASCII

Characters are also stored as integers.

Since the size of a "char" is 1B (8 bits), you can have a unique bidirectional mapping of 2^8 different characters and integers.

The standardised unique mapping scheme is called ASCII

(American Standard code for information interchange)

A - Z	=	65 - 90	"\0" = 0
a - z	=	97 - 122	"\n" = 10
0 - 9	=	48 - 57	" " = 32

Integer Overflow & 2's complements

All data types are stored as Binary Numbers.

Negative integers are stored as 2's complements

2's complement = $\sim x + 1$

(bitwise not, followed by add 1)

Ex. 0101 (5)

↳ 1011 (-5)

2's complement is the established convention to store negative numbers in machines.

Because of the limited size of variables, operations which exceed the range of the variable result in integer overflow (values looping around).

Ex. char a = 55;

a += 256;

// value of char a is still 55

// because it stores only 8 bits (1B)

int b = $2^{31} - 1$;

b += 1;

// now b is -2^{31} because of overflow

Floating Point Explained

sign bit : $\begin{cases} 0 & \text{positive} \\ 1 & \text{negative} \end{cases}$

→ convert number to binary representation

→ shift it till the first "1"
(till after)

→ how many times did you shift?

this is the exponent. store exponent + 127

↳ true exponent

↳ biased exponent

→ the number, clipped after the first "1" is called the mantissa.

Ex. $[-0.087890625]_{10}$

↓
 $[-0.000101101]_2 \rightarrow \text{sign} = 1$

↓
 $\underbrace{0.0001}_\text{clipped} \uparrow \underbrace{01101}_\text{mantissa}$

→ exponent = 4

$$\text{biased exponent} = \text{true exponent} + 127$$

$$= 127 - 4 = 123$$

thus, stored value is (for -0.087890625)

1 0111011 01101000 00000000 0000 000
 exp mantissa

the range of the exponent is -126 to 127
 (1 to 254)

this because there are 2 special values of
 exponent which are reserved.

00000000 $\rightarrow \pm 0$ (0)
 11111111 $\rightarrow \pm \infty$ (255)

comparing values of floats

$\rightarrow$ direct comparison is not recommended
 because the precision is a problem.

Instead we can specify tolerance
 float a, b;

$a == b$ // not recommended

$a - b \leq 0.0000000001$ // recommended

// p.e. $1e-10$ ~~10x~~
 scientific notation

$\text{abs}()$ $\rightarrow$ absolute value, $\text{fabs}()$ abs for float